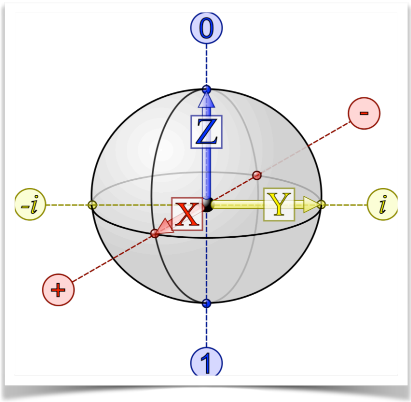


https://elearning.di.unipi.it/course/view.php?id=1043

IQC 2024/25 (756AA, 6CFU)

Introduction to Quantum Computing

Quantum Teleportation

Quantum teleportation protocol

Simple, yet surprising application of elementary quantum mechanics

It involves two parties, **Alice** and **Bob**, who are far away from one another

GOAL

Alice is in possession of **a single qubit, in a quantum state**

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

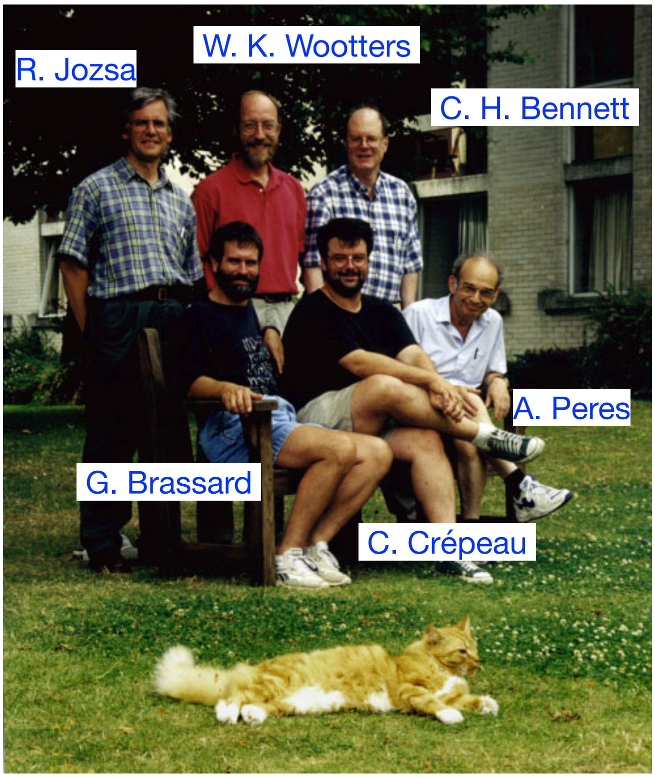
which she would like to send to Bob (very far away)

This protocol was

- discovered and published by C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres, and W. K. Wootters in 1993
- and experimentally demonstrated in 1997

Quantum teleportation can be thought of as the opposite of Superdense coding, in which one transfers two classical bits from Alice to Bob by communicating one qubit

Discoverers of quantum teleportation



Quantum teleportation

- It is the process by which a **QUANTUM STATE is transferred** from one location to another, without any need to directly send the quantum state

- Discovered as a very theoretical result, it has led to
  - important ideas for reducing the effect of noise on quantum computers
  - new hardware ideas for building quantum computers

- We will consider the case of teleporting the simplest possible quantum system: a qubit

Observation

Recall that the NO-CLONING THEOREM states that we cannot make a copy of an arbitrary qubit

When the state of the original qubit is teleported to another location, the state of the original one will be destroyed

**MOVE IS POSSIBLE,
COPY IS IMPOSSIBLE**

Quantum teleportation

Quantum communication protocol

- **Quantum communication channel**

  refer to a communication line (e.g., a fiber-optic cable), which can carry qubits between two remote locations

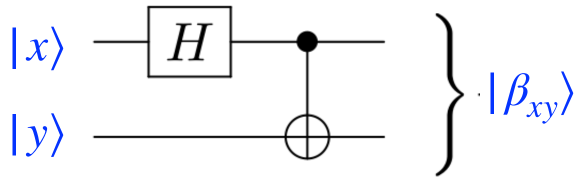
- **Communication channel**

  refer to a channel which can carry classical bits (but not qubits)

The teleportation algorithm works with two entangled qubits, one held by Alice, and one held by Bob
So, Alice and Bob need to share a pair of qubits in an entagled state, e.g, a Bell state

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Bell states



$$|00\rangle \xrightarrow{H\otimes I} \frac{|0\rangle + |1\rangle}{\sqrt{2}}|0\rangle \xrightarrow{CNOT} \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\beta_{00}\rangle$$

$$|01\rangle \xrightarrow{H\otimes I} \frac{|0\rangle + |1\rangle}{\sqrt{2}}|1\rangle \xrightarrow{CNOT} \frac{|01\rangle + |10\rangle}{\sqrt{2}} = |\beta_{01}\rangle$$

$$|10\rangle \xrightarrow{H\otimes I} \frac{|0\rangle - |1\rangle}{\sqrt{2}}|0\rangle \xrightarrow{CNOT} \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\beta_{10}\rangle$$

$$|11\rangle \xrightarrow{H\otimes I} \frac{|0\rangle - |1\rangle}{\sqrt{2}}|1\rangle \xrightarrow{CNOT} \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |\beta_{11}\rangle$$

EVERY VECTOR
IN THIS BASIS IS
ENTANGLED

How to share a pair of entangled qubits?

- The state $|\beta_{00}\rangle$ would have to be created ahead of time, when the qubits are in a lab together and can be made to interact in a way that will give rise to the entanglement between them

- After the state is created, Alice and Bob each take one of the two qubits away with them

ALTERNATIVELY

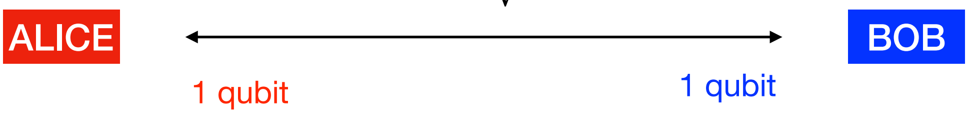
- Some third party may prepare the entangled state ahead of time, sending one qubit to Alice and the other to Bob

- If they are careful not to let them interact with the environment, or other quantum systems, Alice and Bob's joint state will remain entangled

How to share a pair of entangled qubits?

Note that $|\beta_{00}\rangle$ is a fixed state, there is no need for Alice to have sent any qubit to Bob to prepare this state

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



Preliminary

Alice has a single qubit, in a quantum state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Alice would like to send this quantum state to Bob, who is very far away

POSSIBLE APPROACHES

- Alice could physically send the qubit

  (we rule out this possibility because we want to "teleport")

- Alice could tell Bob the amplitudes α and β for the quantum state $|\psi\rangle$

  *To do this, Alice does not need to send a quantum state, but **she could simply send the complex numbers α and β as ordinary classical information** (e.g., over the internet)*

  Bob could then re-create the state in his lab.

In general Alice does not know the identity of her quantum state $|\psi\rangle$

- Alice could measure the state of her qubit

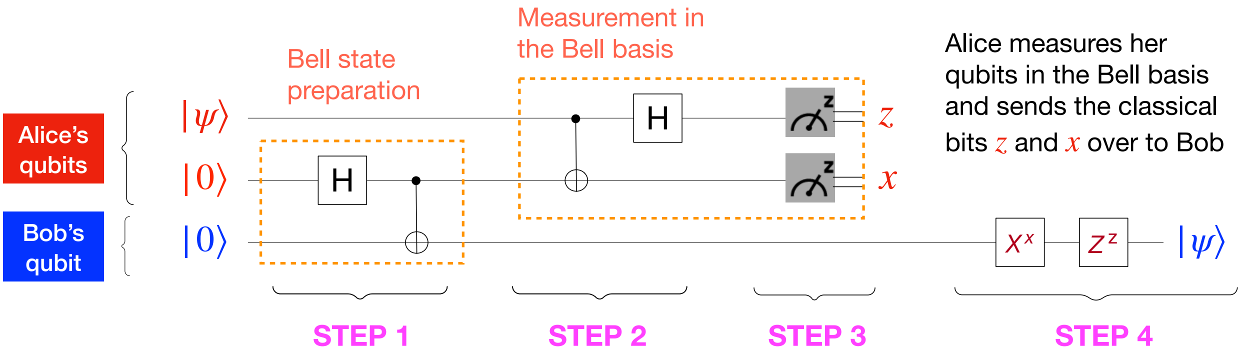
  $$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

  figuring out the amplitudes, and then telling Bob so that he can re-create the state

  - *But the amplitudes α and β are hidden information*
  - *Alice can get only very limited information about α and β, not enough for Bob to re-create $|\psi\rangle$ or a state very close to $|\psi\rangle$*

- Quantum teleportation works even when the identity of the state is not known to Alice and Bob

Quantum circuit for the teleportation protocol



$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$X^1 = X \qquad Z^1 = Z \qquad X^0 = Z^0 = I$$

Initial state (overall state of the three qubits)

$$|\psi\rangle|0\rangle|0\rangle = (\alpha|0\rangle + \beta|1\rangle)|0\rangle|0\rangle = \alpha|000\rangle + \beta|100\rangle$$

State after STEP 1

$$|\psi\rangle|\beta_{00}\rangle$$

The idea behind the protocol

STEP 1

$$|\psi\rangle|\beta_{00}\rangle = \frac{1}{2}\left(|\beta_{00}\rangle|\psi\rangle + |\beta_{01}\rangle\, X|\psi\rangle + |\beta_{10}\rangle\, Z|\psi\rangle + |\beta_{11}\rangle\, XZ|\psi\rangle\right)$$

(EXERCISE: prove it!)

STEP 2

If Alice performs a joint measurement in the Bell basis of the first two qubits ($|\psi\rangle$ and her half of the entangled pair), Bob's qubit collapses in one of the four states

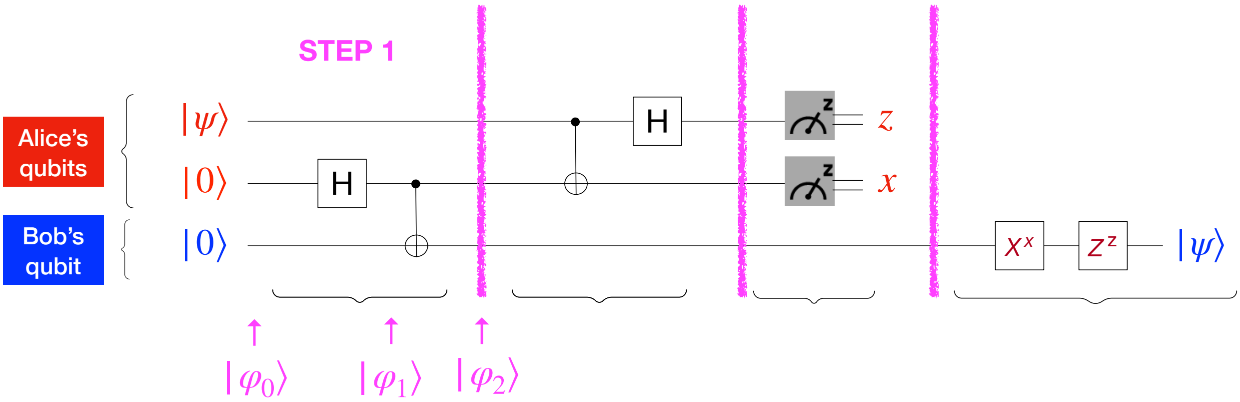
$$|\psi\rangle, \quad X|\psi\rangle, \quad Z|\psi\rangle, \quad XZ|\psi\rangle \qquad \text{(with probability 1/4)}$$

STEP 3 & STEP 4

If Alice sends to Bob the result of the measurement (two classical bits), Bob can perform a final operation to recover $|\psi\rangle$

STEP 1

Two entangled qubits are formed as $|\beta_{00}\rangle$. One is given to Alice, and one to Bob



$$|\varphi_0\rangle = |\psi\rangle|0\rangle|0\rangle$$

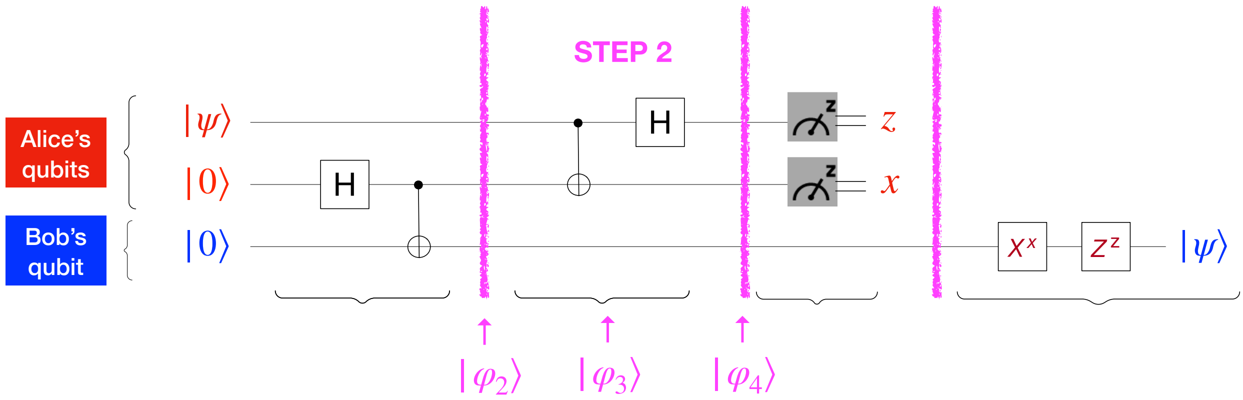
$$|\varphi_1\rangle = (I \otimes H \otimes I)|\varphi_0\rangle = |\psi\rangle\frac{|0\rangle + |1\rangle}{\sqrt{2}}|0\rangle \quad = |\psi\rangle\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$|\varphi_2\rangle = (I \otimes CNOT)|\varphi_1\rangle = |\psi\rangle\frac{|00\rangle + |11\rangle}{\sqrt{2}} = (\alpha|0\rangle + \beta|1\rangle)\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$= \frac{\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)}{\sqrt{2}}$$

STEP 2

Alice lets her qubit $|\psi\rangle$ interact with her entangled qubit



$$|\varphi_2\rangle = \frac{\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)}{\sqrt{2}}$$

$$|\varphi_3\rangle = (CNOT \otimes I)|\varphi_2\rangle \;=\; \frac{\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)}{\sqrt{2}}$$

$$|\varphi_4\rangle = (H \otimes I \otimes I)|\varphi_3\rangle$$

STEP 2

Alice lets her qubit $|\psi\rangle$ interact with her entangled qubit

$$\begin{aligned}|\varphi_4\rangle &= (H \otimes I \otimes I)|\varphi_3\rangle \\ &= (H \otimes I \otimes I)\,\frac{\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)}{\sqrt{2}} \\ &= \frac{\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)}{2} \\ &= \frac{\alpha(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \beta(|010\rangle + |001\rangle - |110\rangle - |101\rangle)}{2}\end{aligned}$$

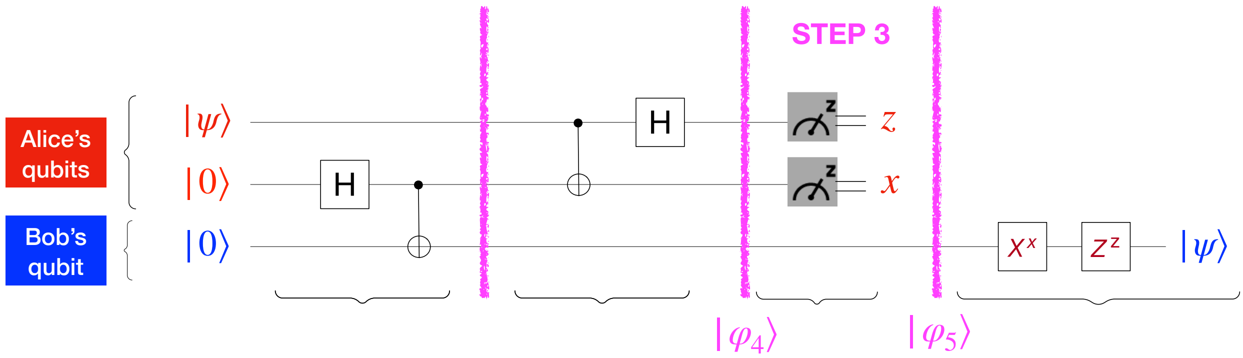
To analyze the measurement in the computational basis of the first two qubits, we group the terms corresponding to each state for those two qubits

$$\begin{aligned}|\varphi_4\rangle = \frac{1}{2}\Big(&|00\rangle\,(\alpha|0\rangle + \beta|1\rangle) \;+\; |01\rangle\,(\alpha|1\rangle + \beta|0\rangle) \\ &+ |10\rangle\,(\alpha|0\rangle - \beta|1\rangle) \;+\; |11\rangle\,(\alpha|1\rangle - \beta|0\rangle)\Big)\end{aligned}$$

The system of three qubits is now in a uniform superposition of 4 states

STEP 3

Alice measures her qubits and determines to which of the possible four states the system collapses



$$|\varphi_4\rangle = \frac{1}{2}\big(|00\rangle\,(\alpha|0\rangle + \beta|1\rangle) \;+\; |01\rangle\,(\alpha|1\rangle + \beta|0\rangle)$$
$$+|10\rangle\,(\alpha|0\rangle - \beta|1\rangle) \;+\; |11\rangle\,(\alpha|1\rangle - \beta|0\rangle)\big)$$

When Alice measures her two qubits, all three qubits collapse to one of four possible states. The outcome of the measurement is

- ➡ 00 with probability $\frac{1}{4}\,(|\alpha|^2 + |\beta|^2) = \frac{1}{4}$

  and the resulting state for Bob's qubits is $|\varphi_5\rangle = \alpha|0\rangle + \beta|1\rangle$

  just the state to be teleported!

- ➡ The other three cases are similar

STEP 3

$$|\varphi_4\rangle = \frac{1}{2}\big(|00\rangle\,(\alpha|0\rangle + \beta|1\rangle) \;+\; |01\rangle\,(\alpha|1\rangle + \beta|0\rangle)$$
$$+|10\rangle\,(\alpha|0\rangle - \beta|1\rangle) \;+\; |11\rangle\,(\alpha|1\rangle - \beta|0\rangle)\big)$$

Measurement outcome $z\,x$	**Probability**	**Bob's state**
00	1/4	$|\varphi_5\rangle = \alpha|0\rangle + \beta|1\rangle = |\psi\rangle$
01	1/4	$|\varphi_5\rangle = \alpha|1\rangle + \beta|0\rangle = X|\psi\rangle$
10	1/4	$|\varphi_5\rangle = \alpha|0\rangle - \beta|1\rangle = Z|\psi\rangle$
11	1/4	$|\varphi_5\rangle = \alpha|1\rangle - \beta|1\rangle = XZ|\psi\rangle$

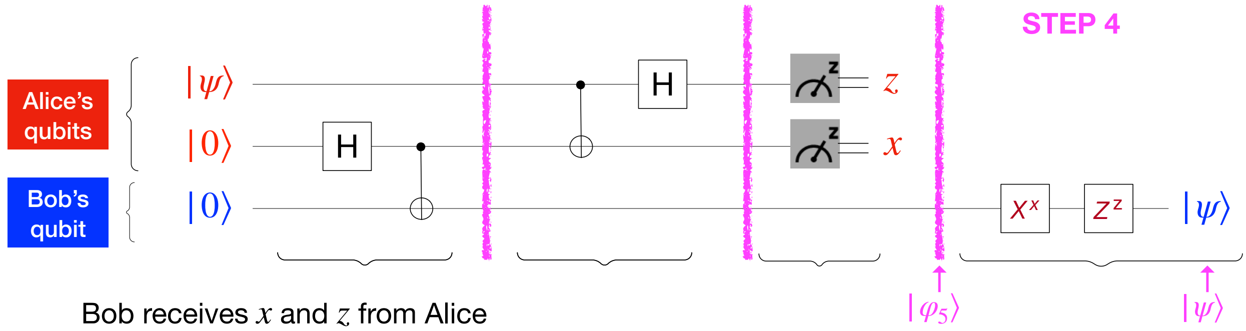
In each case, Bob's state is similar to the original state $|\psi\rangle$ to be teleported.

- Alice knows the state, but Bob does not
- Bob has a state similar to $|\psi\rangle$, but non exactly $|\psi\rangle$

These problems are solved with the last step

STEP 4

Alice sends copies of her two **bits** (not qubits) to Bob
Bob uses this information to achieve the desired state $|\psi\rangle$



Bob receives x and z from Alice

Depending on which state he has (i.e., depending on the bits received), Bob performs the following operation to restore the state $|\psi\rangle$ on his qubit

$z\ \ x$	**Operation**	**Final state**
0 0	$Z^0 X^0 = I$	$\vert\varphi_5\rangle = \alpha\vert 0\rangle + \beta\vert 1\rangle \xrightarrow{I} \alpha\vert 0\rangle + \beta\vert 1\rangle = \vert\psi\rangle$
0 1	$Z^0 X^1 = X$	$\vert\varphi_5\rangle = \alpha\vert 1\rangle + \beta\vert 0\rangle \xrightarrow{X} \alpha\vert 0\rangle + \beta\vert 1\rangle = \vert\psi\rangle$
1 0	$Z^1 X^0 = Z$	$\vert\varphi_5\rangle = \alpha\vert 0\rangle - \beta\vert 1\rangle \xrightarrow{Z} \alpha\vert 0\rangle + \beta\vert 1\rangle = \vert\psi\rangle$
1 1	$Z^1 X^1 = ZX$	$\vert\varphi_5\rangle = \alpha\vert 1\rangle - \beta\vert 0\rangle \xrightarrow{ZX} \alpha\vert 0\rangle + \beta\vert 1\rangle = \vert\psi\rangle$

Bob's qubit is now in the same state $|\psi\rangle$ that Alice had
Alice has successfully teleported her state to Bob

Discussion

To teleport her state to Bob, Alice must send Bob <u>two bits of classical information</u>

- Without these bits, Bob cannot recover $|\psi\rangle$
  (he has only a distribution over four different possible states)
- The speed of the transmission is limited by the speed of light
  (the bits travel along a classical channel)
- Entanglement DOES NOT ALLOW us to communicate faster than the speed of light

It does not matter where Bob is

- Alice may not even know where Bob is
- She can simply broadcast the classical bits out into the world, using a public address on the internet
- Provided that Bob can see the bits, he can recover the state $|\psi\rangle$

At the end of the protocol, Alice is no longer in possession of $|\psi\rangle$, she has only two classical bits

- The measurement leaves her qubits in one of the four states
$$|00\rangle,\ |01\rangle, |10\rangle, |11\rangle$$
corresponding to the result of the measurement

- **Teleportation is a way of moving the state $|\psi\rangle$ rather then copying $|\psi\rangle$**

α and β are complex numbers, s.t. $|\alpha|^2 + |\beta|^2 = 1$, and could have an infinite decimal expansion

Yet, this information has gone from Alice to Bob by passing only two bits

- This information is *"hidden"* and useless to Bob
(as soon as he measures his qubit, it will collapse to a bit)

No particle has been moved at all

- From the point of view of Quantum Mechanics, two particles with the same quantum state are indistinguishable, and can be treated as the same particle

Teleportation is not about an object vanishing in one location and reappearing instantaneously in another

- It is essential that classical information is sent, and Bob performs the corresponding operations

Teleportation is not about transmitting objects over long distances

- It is a way to *disassembling* an unknown state into classical information, and later recovering the original quantum state

There are many variation of the teleportation protocol

- We have seen the simplest version: teleportation of single qubits

- It is possible to extend it to more complex quantum systems, but this seems unlikely in the foreseeable future

- There are also variations that use a different state shared between Alice and Bob, different operations, and so on

<u>EXERCISE</u>: teleport a qubit using the shared state $\|\beta_{01}\rangle$

Conclusion

Teleportation is an astonishing result

- It is impossible to measure a quantum state to determine its amplitudes

  yet

  it is possible to use measurements to teleport a state from one location to another

> Asher Peres, one of the proposers of Quantum Teleportation, was once asked whether it is possible to teleport not only the human body, but also the soul
>
> Peres answered "ONLY THE SOUL"